TECHNICAL CONCEPT PAPER · COMPUTER ARCHITECTURE

Hybrid Photonic Computer Architecture: An Electronic Host and a Photonic Linear-Algebra Accelerator Coupled by Asynchronous Priority Queues

A conceptual system design with base-4 digit-sliced optical encoding, tile-granular preemption, and a parametric energy-per-MAC model.

Alp Eren Celebi

Independent Researcher · Computer Architecture Conceptual Design

Date: October 19, 2026

Version: 1.1 — Revised. Corrects technical claims made in v1.0 (see Appendix A)

Original disclosure: v1.0, March 26, 2026 · DOI 10.17605/OSF.IO/4U6QB

Repository: github.com/alpcelebi/hybrid-photonic-computer

License: MIT — Open for academic use and reference

Status: Conceptual architecture — no hardware or simulation results. All numbers are illustrative estimates with stated assumptions.

ABSTRACT

This paper describes a conceptual hybrid computer in which a conventional binary electronic host offloads linear-algebra workloads (matrix–vector products, convolutions) to a photonic accelerator. The two sides communicate through asynchronous submission and completion queues with three priority classes, held in electronic SRAM, and the accelerator is preempted at tile boundaries so that no optical memory is required.

Rather than encoding 256 optical intensity levels per symbol, which would require roughly 50 dB of signal-to-noise ratio, operands are split into base-4 digits and transmitted as PAM-4 symbols, a format already in production in optical Ethernet. Results are recombined electronically by shift-and-add, using the identity $\mathbf{x} = \Sigma\, 4^k \cdot \mathbf{x}_k$. A parametric energy model, with every assumption stated, puts the photonic core at roughly 100–150 fJ per 8-bit MAC in its best configurations. That is about 5–7× below a contemporary GPU's ~700 fJ/MAC and far short of the 100× often quoted. The model also shows that thermo-optic weight tuning without wavelength multiplexing erases the advantage entirely. Because matrix multiplication is only part of total facility energy, the system-level reduction is bounded at roughly 40% under the assumptions used. Version 1.0 made several claims that this version withdraws or corrects (Appendix A).

Keywords: photonic computing, hybrid electronic–photonic architecture, AI accelerator, PAM-4, digit slicing, thin-film lithium niobate, wavelength division multiplexing, priority queues, preemption, energy per MAC

1. Introduction

Data centres used about 415 TWh of electricity in 2024, roughly 1.5% of global electricity consumption, and demand is growing quickly with AI workloads [15]. Much of the arithmetic in neural-network inference is matrix multiplication, an operation that optical interference can, in principle, perform with an energy that scales favourably with vector size [1, 5, 9].

In practice, photonic accelerators are limited not by the speed of light but by everything around it: the digital-to-analog and analog-to-digital converters at the boundary, the optical signal-to-noise ratio that bounds precision, the power needed to hold weights in tunable elements, and the memory traffic that feeds the accelerator [1, 5]. This paper proposes a system organisation that takes those limits as the starting point. Its contributions are:

1. A system-level arrangement: an unmodified binary electronic host, priority-classed asynchronous queues, and tile-granular preemption, so that the accelerator never needs optical memory (Sections 3 and 7).
2. Base-4 digit slicing, which lowers the optical precision requirement to PAM-4 at the cost of extra passes and makes the number base a design parameter (Section 6).

3. A parametric energy-per-MAC model that identifies which component dominates and bounds the facility-level benefit (Section 8).

No novelty is claimed for any individual component. Each is drawn from published work, cited where it is used.

2. Related Work

Photonic neural-network hardware. Shen et al. demonstrated matrix multiplication with meshes of Mach–Zehnder interferometers [2]. Feldmann et al. used phase-change-material weights and a frequency comb to run parallel convolutions across many wavelengths [4]. Shastri et al. review the field [1], and Pérez-López et al. describe programmable, self-configuring photonic circuits [11]. Ahmed et al. reported an electronic–photonic processor that runs ResNet and BERT models with near-electronic precision for many workloads [6]. That system is the closest prior art to the organisation proposed here.

Energy analyses. Nahmias et al. compared photonic and electronic multiply–accumulate hardware and found the photonic advantage concentrated at large vector sizes and low precision (≈4 bits or less) [5]. Miller analysed the energy limits of optoelectronic conversion [9]. Both results motivate the digit slicing of Section 6 and the model of Section 8.

Materials and devices. Thin-film lithium niobate (TFLN) modulators operate at CMOS-compatible voltages [3, 7]. Silicon nitride offers low-loss waveguides [8]. Multi-level signalling is mature in optical communications [10].

Multi-valued logic and queues. Multi-valued electronic logic has been studied for decades without displacing binary logic in processors [12]. Host–accelerator command queues with doorbells, completion interrupts and priority arbitration are standard. NVMe, for example, specifies weighted round-robin arbitration with an urgent priority class [13]. The queue design in Section 7 adopts those mechanisms.

3. Architecture Overview

The system has three parts. An **electronic host** runs the operating system and applications. A **control die** holds the queues, the converters, the scheduler and the partial-sum memory. A **photonic core** performs linear operations only. All state lives in electronics. The photonic core is stateless between passes.

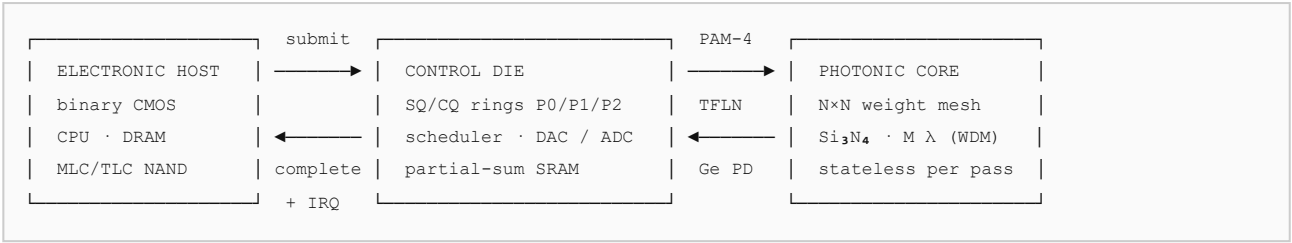


Figure 1. Data flow. The host enqueues work and continues. The control die slices operands into base-4 digits, drives the modulators, digitises the photodetector outputs, recombines partial results and posts completions.

4. Electronic Host

4.1 Why the Host Stays Binary

Version 1.0 proposed base-4 logic with levels at 0, 1.1, 2.2 and 3.3 V. Two problems rule this out. First, at a fixed supply voltage V_{DD}, a binary signal has a gap of V_{DD} between levels while a four-level signal has $V_{DD}/3$, so the noise margin is worse, not better. Second, dynamic power scales as αCV^2f. Raising the supply to 3.3 V from a modern ~0.75 V costs about $(3.3/0.75)^2 \approx 19\times$ in dynamic power.

> Same supply, level gap: binary = V_{DD} quaternary = $V_{DD}/3$
>
> Dynamic power ratio: $(3.3\ V / 0.75\ V)^2 \approx 19\times$
>
> ———————————————————————
>
> Decision: host logic is standard binary CMOS (~0.7–0.9 V)

A binary host also means existing operating systems, compilers and applications run unchanged. The accelerator needs only a driver, a runtime and a compiler back-end, as any GPU or NPU does.

4.2 Where Multi-Level Encoding Belongs

Multi-level encoding works where the physical medium is analog and the extra levels are paid for by slower or more careful sensing. Two examples are storage cells (MLC, TLC and QLC NAND flash, already standard) and optical links (PAM-4). The design uses multi-level encoding in exactly those two places, not in logic gates [12].

5. Photonic Accelerator

5.1 Scope

The photonic core computes linear operations y = Wx, which covers matrix–vector products, convolutions and fixed linear transforms. Non-linear activations, normalisation, control flow and all storage stay in electronics. The design targets inference. Training requires FP16/BF16-class precision for

gradients and weight updates, which is beyond what the optical path provides, so training is out of scope.

5.2 Precision and Signal-to-Noise Ratio

Each additional bit per optical symbol costs about 6 dB of signal-to-noise ratio. The ideal quantisation limit is SNR ≈ 6.02·b + 1.76 dB:

Table 1. Ideal SNR required per optical symbol as a function of bits per symbol.

Bits per symbol	Levels	Required SNR (ideal)	Practicality
2	4 (PAM-4)	13.8 dB	**In production (optical Ethernet)**
4	16	25.8 dB	Demonstrated in research
6	64	37.9 dB	Difficult at tens of GBd
8	256 (v1.0 proposal)	49.9 dB	**Not practical with intensity detection**

Shot noise, laser relative intensity noise and receiver thermal noise all eat into these margins. Intensity is non-negative, so signed values need differential (balanced) detection or offset encoding.

5.3 Speed

Throughput is set by the symbol rate of modulators, converters and detectors, not by the propagation speed of light. Electrical signals in copper also travel at a large fraction of c. TFLN modulators reach tens of GHz of electro-optic bandwidth at CMOS-compatible drive voltages [3, 7]. Time of flight through a millimetre-scale mesh is in the picosecond range, so the latency of a single pass is low, but sustained throughput is symbol rate × parallelism. The model assumes f = 10 GBd per channel, with parallelism from an N×N mesh and M wavelengths.

5.4 Components

Table 2. Photonic-side components and the trade-offs that matter for this design.

Component	Candidate	Trade-off
Waveguides	Silicon nitride (Si_3N_4)	Low propagation loss, broad transparency, CMOS-compatible deposition [8]
Input modulators	Thin-film lithium niobate (TFLN)	Pockels effect, low drive voltage, high bandwidth [3, 7]. Phase shifters are long (mm-scale), which costs footprint
Weight elements	Thermo-optic / electro-optic (TFLN) / non-volatile phase-change [4]	Thermo-optic: mature, but mW-scale static power per element. EO: near-zero static power, larger area. Phase-change: zero hold power, limited endurance
Light source	III–V (InP-based) lasers, external or heterogeneously integrated	Wall-plug efficiency ~10–30%. The rest is heat. Temperature-sensitive, so they need active stabilisation
Photodetectors	Germanium on silicon	Good responsivity at 1310/1550 nm, CMOS-integrable
Converters	PAM-4 drivers (DAC), ~6-bit ADCs on the control die	The ADC dominates conversion energy (Section 8)

6. Base-4 Digit Slicing

6.1 What $4^4 = 256$ Does and Does Not Give

Version 1.0 presented $4^4 = 256$ as a property unique to base 4 that makes the electronic–photonic bridge lossless. It is not unique: $2^8 = 16^2 = 256$ as well, and an 8-bit byte already maps bijectively onto 0–255. Any power-of-two base gives the same exact mapping. The real cost of the bridge is analog conversion (DAC and ADC), which is independent of how the digital bits are grouped.

6.2 Where Base 4 Actually Helps

The identity does become useful when it decides how many optical levels each pass must resolve. Write an 8-bit input as four base-4 digits and use linearity:

$$\mathbf{x} = \sum_k 4^k \cdot \mathbf{x}_k \;,\; \mathbf{x}_k \in \{0,1,2,3\} \implies \mathbf{W}\mathbf{x} = \sum_k 4^k \cdot (\mathbf{W}\,\mathbf{x}_k)$$

Each pass $W \cdot x_k$ needs only PAM-4 inputs (13.8 dB instead of 49.9 dB).
The control die recombines the four partial results by shift-and-add.

The cost is four passes per 8-bit operand, or four parallel wavelength groups. The number base thus becomes a design parameter that trades precision against throughput:

Table 3. Digit-slicing options for an 8-bit operand.

Base	Passes	Optical format	Required SNR (ideal)
2	8	On–off keying	7.8 dB
4 (chosen)	**4**	**PAM-4**	**13.8 dB**
16	2	PAM-16	25.8 dB
256	1	PAM-256	49.9 dB

Base 4 is chosen because PAM-4 transmitters and receivers are already in volume production for optical interconnects, and because it halves the pass count of bit-serial (base-2) operation.

6.3 The Remaining Bottleneck: Output Precision

Slicing the inputs does not reduce the precision needed at the output. Exactly accumulating N = 64 products of 2-bit digits and 4-bit weights needs $\log_2(64 \cdot 3 \cdot 15) \approx 11.5$ bits. ADCs in the energy class assumed in Section 8 deliver about 6 bits at 10 GS/s, so partial sums are quantised. Quantisation-aware training can tolerate this for inference. Otherwise N per tile must shrink, which raises the conversion energy per MAC. This trade-off is the central open problem of the design.

7. Asynchronous Priority Queues

7.1 Where the Queue Lives

Version 1.0 placed the queue in a $LiNbO_3$ crystal. The Pockels effect changes the refractive index only while the field is applied, so $LiNbO_3$ modulates but does not store data. In this version the queues are SRAM ring buffers on the control die (or in host memory, NVMe-style), with doorbell registers for submission and completion queues for results. $LiNbO_3$ moves to where it excels: the input modulators and, optionally, the electro-optic weight elements.

7.2 Priority Classes and Arbitration

Table 4. Priority classes. Arbitration follows the urgent-class + weighted-round-robin scheme of [13].

Class	Use cases	Arbitration
P0 Urgent	Real-time sensing, safety, latency-critical inference	Strict priority. Preempts at the next tile boundary
P1 Normal	Interactive inference, standard matrix work	Weighted round-robin with P2
P2 Background	Batch analytics, offline evaluation	Guaranteed minimum share (e.g. 1 in 16 slots) plus aging, so it never starves

7.3 Tile-Granular Preemption Without Optical Memory

A photonic pass holds no state once it completes. Tile indices, partial sums and digit counters all live in control-die SRAM. The scheduler can therefore switch to a P0 job between any two tiles at negligible cost. The real preemption latency comes from weights: if the P0 job needs different weights, the mesh must be reprogrammed. That takes nanoseconds with electro-optic elements but microseconds or more with thermo-optic ones. Latency-critical work should therefore have its weights kept resident on a dedicated mesh or wavelength group.

7.4 The Non-Blocking Contract, Stated Precisely

Submission never blocks. If a queue is full, the call returns `BUSY` (backpressure) and the host decides whether to retry or yield. Completion is signalled by an interrupt, with coalescing under load. Asynchrony raises throughput by overlapping independent requests, such as batched or pipelined inference. It cannot remove data dependencies: autoregressive decoding, for example, must wait for each token before it can submit the next.

```
submission_entry {
  priority : P0 | P1 | P2
  weight_id : handle to resident weights
  input_ptr : host/SRAM address of operand tile
  output_ptr : address for result tile
  shape : (rows, cols, batch)
  digits : passes per operand (e.g. 4 for 8-bit, base-4)
  timestamp : submission time (for aging)
  cq_id : completion queue to post to
}
```

8. Energy Model

Consider an N×N weight mesh driven at symbol rate f with M wavelengths. Each pass performs $M \cdot N^2$ multiply–accumulates, needs $M \cdot N$ input conversions and $M \cdot N$ output conversions, and burns laser power P_L per wavelength plus static tuning power P_T for the whole mesh. With D passes per operand (digit slicing), the energy per full-precision MAC is:

$$E_{MAC} = D \cdot \left[(E_{DAC} + E_{ADC}) / N + P_L / (f \cdot N^2) + P_T / (f \cdot M \cdot N^2) \right]$$

Conversion cost amortises as 1/N. Laser cost amortises as $1/N^2$. Tuning cost is additionally amortised across wavelengths.

Table 5. Illustrative assumptions. Each value is an order-of-magnitude placeholder to be replaced with device data.

Parameter	Value	Basis
N (mesh size)	64	Limited by output ADC precision (Section 6.3) and optical loss
f (symbol rate)	10 GBd	Well within TFLN modulator bandwidth [3]
D (passes per 8-bit operand)	4	Base-4 digit slicing
E_{DAC}	0.1 pJ	PAM-4 driver
E_{ADC}	1 pJ	~6-bit converter at ~10 GS/s
P_L (wall-plug, per wavelength)	0.3 W	Enough received power for ~6-bit SNR after ~10 dB mesh loss at ~20% laser efficiency
P_T, thermo-optic	8 W	4096 elements × ~2 mW
P_T, electro-optic / non-volatile	0.05 W	Leakage and bias only

Table 6. Model output (8-bit inputs, fJ per MAC). Reference: an NVIDIA H100 SXM delivers ~990 TMAC/s dense FP8 at 700 W, i.e. ~700 fJ/MAC [17].

Configuration	Conversion	Laser	Tuning	Total	vs GPU	Throughput / core
A. Thermo-optic, M = 1	69	29	781	**879**	**0.8× (worse)**	10 TMAC/s
B. Thermo-optic, M = 16	69	29	49	**147**	**4.8×**	164 TMAC/s
C. Electro-optic / NV, M = 1	69	29	5	**103**	**6.9×**	10 TMAC/s
D. Electro-optic / NV, M = 16	69	29	0.3	**98**	**7.2×**	164 TMAC/s

Three conclusions follow:

1. **Weight-element technology decides whether there is any advantage.** Thermo-optic tuning on a single wavelength (A) is worse than the GPU. Either WDM, which amortises tuning (B), or electro-optic / non-volatile weights (C, D) are needed.
2. **The best case is about 5–7×, not 100×.** Once tuning is solved, conversion dominates. Further gains need larger N, which ADC precision and optical loss limit, or cheaper ADCs.
3. **Throughput per core is modest.** At 10–164 TMAC/s per core against ~990 TMAC/s for a GPU, many cores per package are needed, so area and packaging become first-order problems.

These numbers are favourable to photonics. They exclude memory traffic, the host and control-die logic, while the GPU figure includes on-chip data movement.

8.1 Facility-Level Bound

If a fraction φ of facility energy goes to the linear algebra that can be offloaded, and that part becomes S times cheaper, Amdahl's law [16] bounds the total reduction:

$$\text{Reduction} = 1 - [\,(1-\varphi) + \varphi / S\,] \quad \varphi = 0.5,\ S = 7 \Longrightarrow \approx 43\%$$

Memory, networking, host CPUs and facility overheads are unchanged.

Cooling water scales roughly with the heat rejected, so it obeys a similar bound, not the 90% reduction claimed in v1.0. For context, Li et al. estimate that a GPT-3-class model consumes about 500 ml of water per roughly 10–50 medium-length responses, depending on location and season [14].

9. Revised Comparison

Table 7. Contemporary GPU baseline vs this proposal (illustrative, from Section 8).

Aspect	GPU baseline	This proposal	Note
Host logic	Binary CMOS	Binary CMOS (unchanged)	Existing software runs as-is
Storage	MLC/TLC/QLC NAND	Same	Multi-level storage is already standard
Optical symbol	—	PAM-4, digit-sliced to 8 bits	4 passes per operand
Core energy per MAC	~700 fJ (FP8)	~100–150 fJ (best cases)	**~5–7×, memory excluded**
Throughput	~990 TMAC/s per GPU	10–164 TMAC/s per core	**Many cores needed**
Memory bandwidth	3.35 TB/s HBM3 [17]	Set by host memory	Memory wall unchanged
Accumulation precision	FP8–FP32	~6-bit ADC	**Inference-oriented**
Facility energy	Baseline	≤ ~40% lower (φ = 0.5)	Amdahl-bounded
Maturity	Mass production	Research / early commercial [6]	**Integration and packaging risk**

10. Space Applications (Speculative)

Two properties may be useful on spacecraft. First, dielectric waveguides do not suffer the charge-deposition upsets that flip bits in transistor storage. Second, a lower energy cost for linear algebra helps power-limited missions. Both are partial advantages only. Photodetectors, lasers, converters, the control die and the host are still semiconductor electronics and need radiation hardening. Optical materials also suffer radiation-induced attenuation, and lasers need thermal stabilisation. Version 1.0's claims of radiation immunity, a 5,000× capability gain and terabit deep-space links are withdrawn. Deep-space optical links are limited by received photons (link budget), not by the number of wavelength channels. Any space use would require radiation testing of the full optoelectronic stack.

11. Limitations and Open Problems

1. **Output precision.** ~6-bit ADCs quantise the partial sums (Section 6.3). Larger tiles reduce energy per MAC but need more ADC bits.
2. **Weight reprogramming.** Thermo-optic elements are slow and power-hungry. Electro-optic elements are large. Phase-change elements have limited endurance. Weight-stationary operation with batching is assumed.
3. **Optical loss scaling.** Insertion loss grows with mesh depth, which raises the laser power needed for a given SNR.
4. **Integration and packaging.** The photonic core, the control die and the lasers must be co-packaged with dense electrical and optical I/O.
5. **Lasers.** Wall-plug efficiency of 10–30% and temperature sensitivity add heat and require control loops.
6. **Memory wall.** The accelerator does not reduce weight or activation traffic from host memory, which dominates many inference workloads.
7. **Training.** Out of scope at the available precision.
8. **Software.** A compiler back-end must map models onto tiles, digit slices and resident weights. The host software stack is unchanged.
9. **No validation yet.** The model in Section 8 has not been checked against device-level simulation or measurements.

12. Conclusion

This revision recasts the hybrid photonic computer as a binary electronic host with a photonic linear-algebra accelerator, coupled by priority-classed asynchronous queues and preempted at tile boundaries so that no optical memory is needed. The $4^4 = 256$ relation survives as a design choice: base-4 digit slicing lets each optical pass use mature PAM-4 signalling.

The energy model gives a sober outlook. A best-case core-level gain of about 5–7× over a contemporary GPU is plausible. That gain depends on electro-optic or non-volatile weights, or on wavelength multiplexing, and it shrinks at the facility level because of Amdahl's law. The most valuable next steps are, first, replacing the placeholder parameters of Table 5 with measured device data and, second, a circuit-level study of the ADC precision–tile size trade-off.

The author welcomes correspondence from researchers working on related problems.

Appendix A. Errata Relative to v1.0

Table A1. Claims in v1.0 (March 26, 2026) that are corrected or withdrawn in this version.

#	v1.0 claim	Correction
1	Only base 4 maps exactly to 256 levels, making the bridge lossless	Every power-of-two base maps exactly ($2^8 = 16^2 = 256$). The real cost is DAC/ADC conversion. Base 4 is retained for digit slicing (Section 6)
2	"Octal-equivalent" photonic processing	8 bits per symbol is not octal (base 8). Wording removed
3	Base-4 logic at 0/1.1/2.2/3.3 V with noise margins exceeding binary	At a fixed supply the four-level gap is $V_{DD}/3$. 3.3 V costs ~19× dynamic power. The host is binary CMOS (Section 4)
4	256 intensity levels per optical symbol	Requires ~50 dB SNR. Replaced by PAM-4 with digit slicing (Sections 5.2, 6)
5	Terahertz processing, 100–1000× faster	Symbol rates are tens of GBd. Throughput comes from parallelism (Section 5.3). v1.0's own 10 Tbps figure implied ~16 GBd, not THz
6	Photons produce no heat, so cooling is nearly eliminated	Lasers, tuning and converters dissipate heat. Best-case core gain is ~5–7×. Facility reduction ≤ ~40% (Section 8)
7	The $LiNbO_3$ crystal stores the priority queue	The Pockels effect is not memory. Queues are in SRAM and $LiNbO_3$ is used as the modulator (Section 7.1)
8	P0 preempts photonic tasks (implicitly requiring optical state storage)	Tile-granular preemption with all state in electronics (Section 7.3)
9	Neither system ever waits for the other	Submission is non-blocking with backpressure. Data dependencies still serialise work (Section 7.4)
10	Table 3 projections: 90% less water, 95% less energy, GPT-4 training in 3–7 days, <10 ms inference, ~100 Gbps bandwidth today	Withdrawn as unsupported and replaced by the model of Section 8. A current GPU already has 3.35 TB/s of memory bandwidth [17]
11	AI accounts for 3–4% of global emissions, exceeding aviation	Data centres as a whole used ~1.5% of global electricity in 2024 [15]. AI is a subset of that

12	Space: cosmic-ray and solar-storm immunity, 5,000× capability, >10 Tbps deep-space links	Withdrawn. Partial potential advantages only (Section 10)
13	A new OS, compilers and a Base-4 ISA are required	Not required with a binary host. Only an accelerator runtime and compiler back-end are needed
14	References: an unrelated laser-processing review; a 2004 MRS Bulletin article on silicon nitride that could not be located; an unverified vendor note	Removed and replaced with verifiable sources

References

[1] Shastri, B. J., Tait, A. N., Ferreira de Lima, T., Pernice, W. H. P., Bhaskaran, H., Wright, C. D., & Prucnal, P. R. (2021). Photonics for artificial intelligence and neuromorphic computing. *Nature Photonics*, 15, 102–114.

[2] Shen, Y., Harris, N. C., Skirlo, S., Prabhu, M., Baehr-Jones, T., Hochberg, M., … & Soljačić, M. (2017). Deep learning with coherent nanophotonic circuits. *Nature Photonics*, 11(7), 441–446.

[3] Wang, C., Zhang, M., Chen, X., Bertrand, M., Shams-Ansari, A., Chandrasekhar, S., … & Lončar, M. (2018). Integrated lithium niobate electro-optic modulators operating at CMOS-compatible voltages. *Nature*, 562(7725), 101–104.

[4] Feldmann, J., Youngblood, N., Karpov, M., Gehring, H., Li, X., Stappers, M., … & Pernice, W. H. P. (2021). Parallel convolutional processing using an integrated photonic tensor core. *Nature*, 589(7840), 52–58.

[5] Nahmias, M. A., Ferreira de Lima, T., Tait, A. N., Peng, H.-T., Shastri, B. J., & Prucnal, P. R. (2020). Photonic multiply-accumulate operations for neural networks. *IEEE Journal of Selected Topics in Quantum Electronics*, 26(1), 1–18.

[6] Ahmed, S. R., et al. (2025). Universal photonic artificial intelligence acceleration. *Nature*, 640(8058), 368–374. doi:10.1038/s41586-025-08854-x

[7] Zhu, D., Shao, L., Yu, M., Cheng, R., Desiatov, B., Xin, C. J., … & Lončar, M. (2021). Integrated photonics on thin-film lithium niobate. *Advances in Optics and Photonics*, 13(2), 242–352.

[8] Blumenthal, D. J., Heideman, R., Geuzebroek, D., Leinse, A., & Roeloffzen, C. (2018). Silicon nitride in silicon photonics. *Proceedings of the IEEE*, 106(12), 2209–2231.

[9] Miller, D. A. B. (2017). Attojoule optoelectronics for low-energy information processing and communications. *Journal of Lightwave Technology*, 35(3), 346–396.

[10] Agrell, E., Karlsson, M., Chraplyvy, A. R., Richardson, D. J., Krummrich, P. M., Winzer, P., … & Willner, A. E. (2016). Roadmap of optical communications. *Journal of Optics*, 18(6), 063002.

[11] Pérez-López, D., López, A., DasMahapatra, P., & Capmany, J. (2020). Multipurpose self-configuration of programmable photonic circuits. *Nature Communications*, 11(1), 6359.

[12] Hurst, S. L. (1984). Multiple-valued logic — its status and its future. *IEEE Transactions on Computers*, C-33(12), 1160–1179.

[13] NVM Express, Inc. (2021). *NVM Express Base Specification*, Revision 2.0 (command arbitration: round robin and weighted round robin with urgent priority class).

[14] Li, P., Yang, J., Islam, M. A., & Ren, S. (2023). Making AI less "thirsty": Uncovering and addressing the secret water footprint of AI models. *arXiv:2304.03271*.

[15] International Energy Agency (2025). *Energy and AI*. IEA, Paris.

[16] Patterson, D. A., & Hennessy, J. L. (2013). *Computer Organization and Design: The Hardware/Software Interface* (5th ed.). Morgan Kaufmann.

[17] NVIDIA Corporation (2023). *NVIDIA H100 Tensor Core GPU Datasheet*.